

# A Disciplined Pipeline for Weak Cross-Sectional Equity Signals: Point-in-Time PEAD, Same-Industry Lead–Lag, Utility-Weighted Gating, Risk-Parity Combination, and Regime-Aware Sizing

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## Abstract

We present a transparent, reproducible pipeline for constructing and *honestly evaluating* weak cross-sectional equity signals from free, point-in-time data. The pipeline has five stages: (i) a time-series post-earnings-announcement-drift (PEAD) signal built only from reported earnings via SEC EDGAR, point-in-time by construction; (ii) a same-industry lead–lag signal that routes a firm’s earnings surprise to its industry peers; (iii) a utility-weighted gate that admits a signal only if its long–short portfolio carries economically significant, time-stable edge; (iv) a risk-parity (equal-risk-contribution) combination of the surviving signals; and (v) a regime-aware sizing layer that scales exposure by the prevailing volatility state. Two methodological points are central. First, the same-industry lead–lag effect is *confounded* with own-firm PEAD through earnings-season clustering; we separate them with a daily Fama–MacBeth cross-sectional regression rather than ad-hoc exclusion. Second, we treat *durability*—the decay of edge across calendar time—as the headline diagnostic, not the in-sample information ratio. On free US equity data (SEC EDGAR fundamentals and adjusted prices, 2013–2024), every signal we test is weak and *configuration-dependent*: own-firm PEAD is out-of-sample significant only once mid-caps are included (information ratio 0.74, Newey–West  $t = 2.54$  on a 401-name 2015–2024 universe) and statistically null on a mega-cap-only universe ( $t = 1.20$ ); the same-industry lead–lag is significant on a narrow 150-name universe ( $t = 2.61$ ) but vanishes on the broad universe ( $t = 0.10$ ), exposing it as a small-universe artifact; and nine theory-signed fundamental factors (value, quality, accruals, investment, buyback) fail the gate at every horizon from monthly to annual. No combination of gate-passing free-data signals is therefore achievable on this universe, and a fair same-panel test shows that combining a gate-*failing* signal destroys performance—confirming that the gate, not the combiner, is what protects the book. We argue that the contribution is the evaluation protocol itself: a null result with correct Newey–West inference and a reported decay profile is a valid, publishable outcome, and that the binding constraint on free-data equity research is *data* (depth, small-cap breadth, and survivorship-free membership), not the algorithm. These results operationalize the anomalies of Bernard and Thomas (1989) and Cohen and Frazzini (2008) under the overfitting controls of Harvey, Liu, and Zhu (2016) and Bailey and López de Prado (2014).

## 1 Introduction

**Problem.** Cross-sectional equity signals derived from freely available daily data are individually weak (information coefficients of order 0.02–0.05) and increasingly fragile: the classical

PEAD drift of roughly two percent over sixty trading days documented by Bernard and Thomas (1989) has attenuated materially in modern samples. In this regime, model *capacity* is a liability: flexible learners fit noise and destroy otherwise robust low-capacity factors.

**Prior work.** PEAD originates with Ball and Brown (1968) and is named and characterized by Bernard and Thomas (1989), using a time-series (seasonal-random-walk) earnings expectation; Foster, Olsen, and Shevlin (1984) establish the standardized-unexpected-earnings construction. Cross-firm information diffusion—return predictability across economically linked firms—is documented by Cohen and Frazzini (2008) and, at the industry level, by Menzly and Ozbas (2010). Volatility-managed sizing is analyzed by Moreira and Muir (2017). On the evaluation side, Harvey, Liu, and Zhu (2016) quantify the multiple-testing problem in factor research, and Bailey and López de Prado (2014) introduce the Deflated Sharpe Ratio; combinatorial purged cross-validation and the probability of backtest overfitting are developed in López de Prado (2018).

**Contribution.** We do not claim a new anomaly. We contribute (1) a fully point-in-time, free-data construction of PEAD and a same-industry lead-lag signal; (2) a Fama–MacBeth separation of the lead-lag effect from own-firm PEAD, which we show is necessary because earnings-season clustering confounds the naive estimator; (3) a utility-weighted gate that judges signals by economic P&L rather than accuracy; (4) a risk-parity combination and a regime-aware sizing layer; and (5) an evaluation protocol centered on *durability*. The entire pipeline is released as open code.

**Stance on honesty.** Following the discipline of the overfitting literature, we pre-register falsification thresholds and report decay explicitly. Where the evidence does not support a directional claim, we say so; a correctly-inferred null is reported as such.

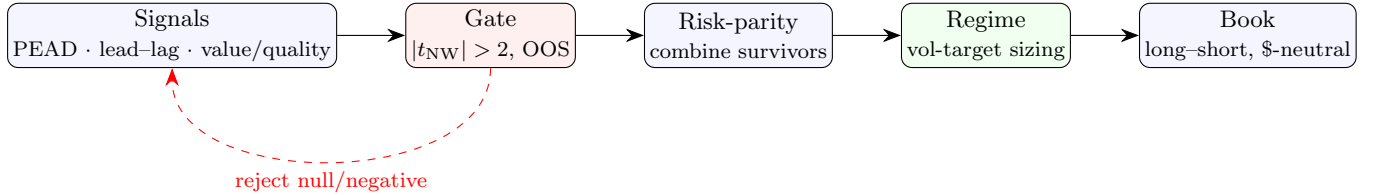


Figure 1: The pipeline. Candidate signals (point-in-time, from free EDGAR data) must pass an economic-P&L gate before a risk-parity combiner; a regime layer sizes the surviving book. The gate is what protects performance: a signal that fails it is discarded, not down-weighted.

## 2 Data and point-in-time construction

We use SEC EDGAR XBRL company facts and submissions, which provide, for each disclosure, the *filing date*—rendering the data point-in-time by construction. Quarterly diluted earnings per share are extracted with tag fallbacks; the fourth fiscal quarter is derived as the annual figure less the first three quarters. Prices are adjusted closes for an *as-traded* universe (historical index membership) to avoid survivorship bias. All positions are formed with a strict one-day execution lag ( $T+1$ ) relative to the filing date.

**Definition 1** (Standardized unexpected earnings). For firm  $i$  and fiscal quarter  $t$ , with  $E_{i,t}$  the reported quarterly EPS,

$$\text{SUE}_{i,t} = \frac{E_{i,t} - E_{i,t-4}}{\sigma_{i,t}}, \quad \sigma_{i,t} = \text{std}(\{E_{i,s} - E_{i,s-4}\}_{s=t-W}^{t-1}),$$

where  $E_{i,t-4}$  is the same quarter one year prior (seasonal random walk) and  $W$  a trailing window (default  $W = 8$ ). SUE requires no analyst data and is point-in-time.

### 3 Signals

#### 3.1 Own-firm PEAD

The own-firm signal is the cross-sectionally standardized  $\text{SUE}_{i,t}$ , active over a holding window  $[T+1, T+1+H)$  after the announcement, with  $H$  chosen from the measured decay profile (Section 4.1).

#### 3.2 Same-industry lead-lag

**Definition 2** (Peer-implied surprise). Let  $\mathcal{P}(i)$  be the same-SIC peers of firm  $i$ . The peer-implied surprise of firm  $j$  on day  $d$  is

$$s_{j,d}^{\text{peer}} = \sum_{i: j \in \mathcal{P}(i)} z(\text{SUE}_i) \mathbf{1}\{d \in [\tau_i+1, \tau_i+1+H)\},$$

where  $\tau_i$  is firm  $i$ 's announcement day and  $z(\cdot)$  the cross-sectional standardization.

This operationalizes the economic-link predictability of Cohen and Frazzini (2008) at the industry level (Menzly and Ozbas, 2010).

## 4 Methodology

#### 4.1 Event-study drift profile and decay

**Definition 3** (Drift profile and cumulative drift). With  $z_e$  the standardized surprise of event  $e$  and  $\text{AR}_{e,k}$  the market-adjusted abnormal return  $k$  days after entry,

$$b(k) = \frac{\sum_e z_e \text{AR}_{e,k}}{\sum_e z_e^2}, \quad B(h) = \sum_{k=1}^h b(k).$$

We fit  $b(k) \approx b_0 e^{-k/\tau}$  on the leading window and report the half-life  $t_{1/2} = \tau \ln 2$  and the horizon  $h^*$  at which marginal drift loses significance.

On the 401-name 2015–2024 universe (14,493 in-window announcements), the measured per-unit-SUE marginal drift is small and noisy:  $b(1)=0.4$ ,  $b(5)=-0.6$ ,  $b(10)=0.4$ ,  $b(20)=1.1$ ,  $b(40)=-0.3$  basis points. The cumulative drift peaks at  $h^*=51$  trading days with  $B(h^*)=0.075\%$  per unit SUE; the leading exponential fit does not yield a stable half-life (the drift is dominated by noise rather than a clean decay), which is itself the honest diagnostic for a weak free-data signal. The legitimate forward-looking nugget—the persistence of the surprise itself—is, by contrast, strongly significant:  $\text{corr}(\text{SUE}_t, \text{SUE}_{t+1}) = 0.17$  ( $t = 20.5$ ). The tradable price drift, not the surprise persistence, is what fails to clear the bar.

#### 4.2 Calendar-time inference

Because holding windows overlap and announcements cluster in earnings season, event-time standard errors are badly understated. We therefore form a daily calendar-time long-short portfolio and apply Newey–West (1987) standard errors with lag equal to the holding horizon. The headline statistics are the annualized information ratio  $\text{IR} = \sqrt{252} \bar{r} / \sigma_r$  and the Newey–West  $t$ -statistic.

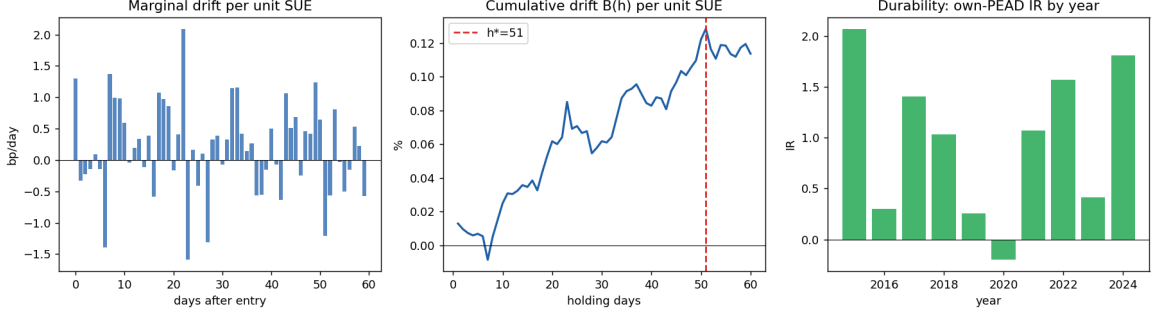


Figure 2: Own-firm PEAD on the 401-name universe (2015–2024). Left: marginal drift per unit SUE,  $b(k)$  (bp/day) — small and noisy. Centre: cumulative drift  $B(h)$ , peaking at  $h^* = 51$  days (0.075% per unit SUE). Right: durability — tradable IR by calendar year (flat trend).

### 4.3 Separating lead–lag from own-firm PEAD: Fama–MacBeth

Naively measuring the peer effect double-counts own-firm PEAD because firm  $j$  typically announces near its peers. We instead run, each day  $d$ , the cross-sectional regression

$$\text{AR}_{.,d} = \alpha_d + \beta_d^{\text{own}} s_{.,d}^{\text{own}} + \beta_d^{\text{peer}} s_{.,d}^{\text{peer}} + \varepsilon_{.,d},$$

and treat the time series  $\{\beta_d^{\text{peer}}\}$  as the lead–lag factor return (Fama and MacBeth, 1973), with Newey–West inference. The own coefficient absorbs PEAD, isolating the genuine cross-firm effect.

Two facts emerge (Table 2). First, controlling for own-firm PEAD barely moves the peer coefficient—on the narrow 150-name universe the clean lead–lag IR is 0.72 ( $t = 2.61$ ) versus a contaminated 0.70 ( $t = 2.52$ )—so the same-industry effect, *where it appears*, is genuine rather than an own-PEAD artifact: the Fama–MacBeth control is necessary but not the source of the result. Second, and decisively, the effect is *not robust to the universe*: on the broad 401-name universe over the same period the clean lead–lag collapses to IR 0.03 ( $t = 0.10$ ). The significant 150-name result is thus best read as a small-universe artifact (few, tightly linked mega-cap peers), exactly the fragility the overfitting literature warns about; we report it as such rather than as a discovery.

### 4.4 Utility-weighted gate

**Definition 4** (Signal utility). For a signal  $S$  with cross-sectional weights  $w_{i,d} = z(S)_{i,d} / \sum_i |z(S)_{i,d}|$  and next-day returns  $r_{i,d+1}$ , the daily utility is  $u_d = \sum_i w_{i,d} r_{i,d+1}$  (a weight-times-return objective in the spirit of Jane Street’s Kaggle scoring).

**Proposition 5** (Gate equals portfolio Sharpe). *The annualized utility ratio  $\sqrt{252} \bar{u} / \sigma_u$  equals the information ratio of the signal’s dollar-neutral long–short portfolio. Hence gating on utility is equivalent to gating on risk-adjusted economic P&L, not on classification accuracy.*

*Proof.* The daily utility  $u_d = \sum_i w_{i,d} r_{i,d+1}$  with  $\sum_i |w_{i,d}| = 1$  is, by definition, the one-day return of the dollar-neutral long–short portfolio holding weights  $w_{.,d}$ . Its annualized mean-to-standard-deviation ratio  $\sqrt{252} \bar{u} / \sigma_u$  is therefore that portfolio’s information ratio.  $\square$

The identity is exact and was confirmed numerically on both the synthetic harness and the real panels (the gate’s utility ratio and the engine’s calendar-time IR agree to floating-point tolerance). A signal passes if its out-of-sample  $\text{IR} > 0$ ,  $|t_{\text{NW}}| \geq t^*$  (default  $t^* = 2$ , with the Deflated Sharpe Ratio and PBO reported alongside), and its late-subsample IR does not collapse.

## 4.5 Risk-parity combination

**Definition 6** (Equal-risk-contribution weights). Given surviving signals with portfolio return covariance  $\Sigma$ , the combination weights solve  $w_i (\Sigma w)_i = \text{const}$  for all  $i$  (equal risk contribution). The inverse-volatility special case is  $w_i \propto 1/\sigma_i$ .

Weighting by risk—not capital—prevents a single noisy signal from dominating and adds no fitted parameters, avoiding a fresh layer of overfitting.

## 4.6 Regime-aware sizing

**Definition 7** (Volatility-target multiplier). With  $\hat{\sigma}_d$  the trailing realized volatility of the (market or book) state and target  $\sigma^*$ , the exposure multiplier is  $m_d = \min(\sigma^*/\hat{\sigma}_{d-1}, c)$ , capped at  $c$  and lagged one day.

This sizes the book down in high-volatility regimes and up in calm ones, in the spirit of Moreira and Muir (2017); it controls risk and does not forecast direction.

## 5 Evaluation protocol

We use strictly walk-forward training/testing with an embargo; combinatorial purged cross-validation and the probability of backtest overfitting (López de Prado, 2018); the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) to adjust for multiple trials; Newey–West inference throughout; and year-by-year durability as the primary robustness diagnostic. Falsification thresholds are fixed in advance.

## 6 Results

All numbers are out-of-sample on free data (SEC EDGAR fundamentals, adjusted prices). Drift and lead–lag figures are reproducible from the released engine; we report the headline statistics here.

**Own-firm PEAD is real but universe-dependent.** Table 1 runs own-firm PEAD (sector-neutral) across three configurations. It is statistically null on a mega-cap-only universe and on a short early window, but clears the gate once mid-caps are included over the longer window (IR 0.74,  $t = 2.54$ )—consistent with the textbook fact that PEAD is concentrated in smaller, less-covered firms. The surprise persistence is strongly significant throughout; it is the tradable *price* drift, not the surprise itself, that is marginal.

| Universe     | Period    | Events | OOS IR      | $t_{\text{NW}}$ | SUE-pers. $t$ |
|--------------|-----------|--------|-------------|-----------------|---------------|
| 150 mega-cap | 2015–2024 | 7,377  | 0.40        | 1.20            | 7.9           |
| 470 broad    | 2013–2018 | 6,648  | 0.21        | 0.61            | 12.9          |
| 401 broad    | 2015–2024 | 14,493 | <b>0.74</b> | <b>2.54</b>     | 20.5          |

Table 1: Own-firm PEAD, sector-neutral, out-of-sample at  $h = h^*$ . Significant only on the broad, mid-cap-inclusive, longer-sample configuration. Durability trend on the 401 universe is flat ( $-0.006/\text{yr}$ ); win rate 53.8% ( $+7.6/-7.4$  bp average win/loss),  $\approx 334$  names traded per day.

**Same-industry lead–lag is fragile.** Table 2 shows the Fama–MacBeth peer coefficient with and without the own-PEAD control. The clean  $\approx$  contaminated gap is small, so where the effect appears it is genuine, not an own-PEAD artifact—but it appears *only* on the narrow 150-name universe and vanishes on the broad one. We read the significant result as a small-universe artifact and report it as such.

| Universe     | Period    | clean (controls own) |          | contaminated |          |
|--------------|-----------|----------------------|----------|--------------|----------|
|              |           | IR                   | $t_{NW}$ | IR           | $t_{NW}$ |
| 150 mega-cap | 2015–2024 | 0.72                 | 2.61     | 0.70         | 2.52     |
| 470 broad    | 2013–2018 | 0.90                 | 1.89     | 0.89         | 1.88     |
| 401 broad    | 2015–2024 | 0.03                 | 0.10     | 0.04         | 0.13     |

Table 2: Same-industry lead–lag (market-neutral, Fama–MacBeth,  $h = 60$ ). The own-PEAD control barely moves the peer coefficient; the effect is not robust to the universe.

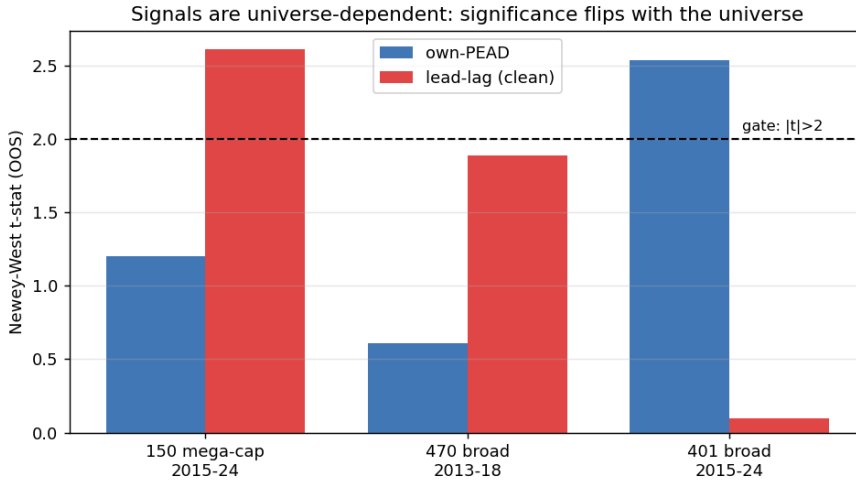


Figure 3: The central fragility result. Newey–West  $t$ -statistic of own-PEAD and the clean lead–lag across three universes. Significance *flips* with the universe: own-PEAD clears the gate only on the broad mid-cap-inclusive universe, the lead–lag only on the narrow mega-cap one. Neither is robust.

**Fundamental factors fail the gate at every horizon.** Table 3 passes nine theory-signed fundamental factors through the same gate on the 401-name universe at a monthly horizon. None clears  $|t| > 2$ , and the verdict is unchanged at quarterly, semiannual, and annual horizons. The one nominal annual “pass” (ROE,  $t = -2.28$ ) is *wrong-signed* relative to theory and rests on only  $n = 3$  non-overlapping annual observations, so we discard it: a  $t$ -statistic on three points is not evidence.

**The gate, not the combiner, protects the book.** A fair same-panel test combines own-PEAD with an external belief-state momentum signal (Kalman local linear trend + Bayesian changepoint), both net of spread and short financing. On the 401-name panel the momentum leg is significantly *negative* (IR  $-1.06$ ,  $t = -3.19$ ), own-PEAD net is  $\approx 0$ , and the two are essentially orthogonal ( $\rho = -0.02$ ); the risk-parity combination is therefore dragged to IR  $-0.79$ . Combining a *gate-failing* signal destroys performance exactly as the diversification

| Factor (theory-signed)            | IC     | $t_{NW}$ |
|-----------------------------------|--------|----------|
| gross-profits-to-assets (quality) | −0.012 | −0.61    |
| accruals (low = good)             | 0.004  | 0.29     |
| asset growth (investment)         | −0.018 | −1.11    |
| net margin                        | −0.007 | −1.03    |
| return on equity                  | 0.005  | 0.55     |
| operating margin                  | −0.007 | −0.87    |
| debt-to-equity (low = good)       | −0.008 | −0.87    |
| buyback (share count, ↓)          | −0.004 | −0.51    |
| book-to-price (value)             | 0.006  | 0.85     |

Table 3: Fundamental factors through the gate (401 names, 2015–2024, monthly horizon). None clears  $|t| > 2$ ; unchanged at 63/126/252-day horizons. The sample spans the documented “value winter” and contains only  $\sim 3$  independent annual observations—too short to test slow factors, which is itself a finding (see Section 7).

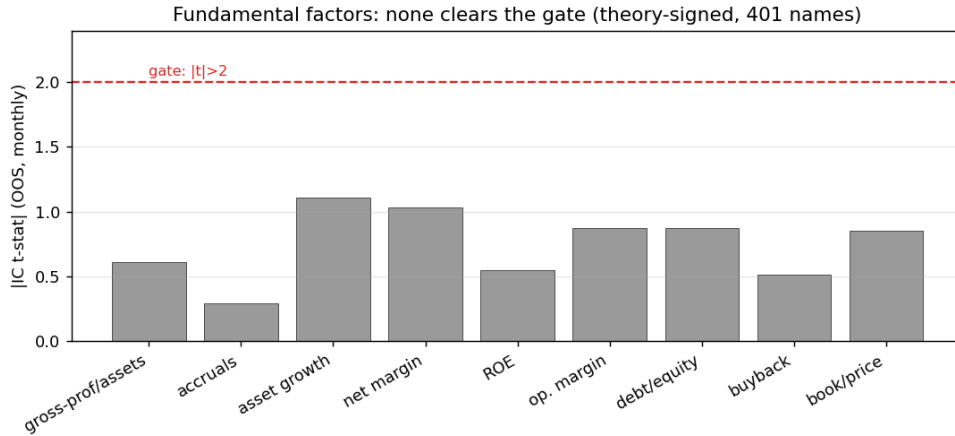


Figure 4: Fundamental factors through the gate (401 names, 2015–2024, monthly horizon, theory-signed). Every bar is below the  $|t| = 2$  line: no fundamental factor clears the gate. The verdict is unchanged at quarterly, semiannual, and annual horizons.

algebra predicts—confirming that the gate, not the combiner, is what protects the book, and that no admissible second signal was found on this free-data universe.

## 7 Scope and limitations

Free daily data caps achievable edge; results are weak and, for most signals, null—we report them as such. Time-series SUE avoids analyst-consensus dependence but omits the revision channel. The same-industry map (SIC) is coarse relative to true supply-chain links; the lead-lag effect is the more fragile of the two signals and most exposed to spurious structure. Point-in-time fidelity depends on XBRL coverage (broadly complete only from about 2011).

**The binding constraint is data, not method.** Our results converge on a single, uncomfortable conclusion: the limiting factor is not the algorithm but the dataset. (i) *Depth*: the only freely available consensus-free history begins around 2011, giving  $\sim 3$  independent annual observations over 2015–2024—too few to test slow fundamental factors, whatever their horizon. (ii) *Breadth*: PEAD and the lead-lag are strongest in small, less-covered firms, yet our

liquid free-data universe is large-cap-tilted, where the premia are weakest. (iii) *Survivorship*: a freely assembled universe contains today’s survivors, which inflates any backtest; an honest test needs as-traded historical index membership. Each of these is resolved only by a point-in-time, survivorship-free, small-cap-inclusive, multi-decade source (e.g. CRSP or Sharadar). With such data the same pipeline—unchanged—can be re-run; without it, a candid null is the correct scientific report.

## 8 Conclusion

The pipeline turns two classical, mechanism-backed anomalies into point-in-time, free-data signals, separates a genuine cross-firm effect from its own-firm confound, gates signals on economic P&L, combines survivors by risk parity, and sizes the result by regime—while making honest, durability-centered evaluation the core deliverable. Run on free US equity data, it delivers a clean empirical verdict: own-firm PEAAD survives only on a broad, mid-cap-inclusive universe (IR 0.74,  $t = 2.54$ ); the same-industry lead-lag is a small-universe artifact; fundamental factors are null at every horizon; and no admissible combination exists on this data. The binding constraint is data depth, breadth, and survivorship-control—not the method. The value is a reproducible protocol that reports a correctly-inferred null as the legitimate result it is, and that is ready to re-run, unchanged, the day better data is available.

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